

MIMIR: Verifiable, Privacy-Preserving Exam Generation with Local Language Models

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Abstract—The simplest way to obtain an exam from teaching material today is to upload the document to a chat-based large language model (LLM) and ask for a test. This gives no evidence that the keyed answers are supported by the document, no control over which parts of the material are covered, and it sends the material to a third party. We present MIMIR, an open system that generates multiple-choice exams from a teacher’s own documents with a small language model running on a consumer GPU. MIMIR replaces the single prompt by an explicit method: semantic chunking, a *blueprint* that assigns concepts, question types and Bloom levels to slots, per-question generation with chunk citations, and a verifier chain that checks grounding, distractors, blind answerability and answer leakage before a mandatory human review. Uploaded content is kept only for the duration of a job. We also contribute an evaluation methodology: fourteen single-factor arms, a 39-document Hungarian and English dataset, a cross-family LLM judge and document-level statistics. A pilot exposed a malformed fallback and an answer-leakage failure mode that fooled the verifier and the judge. In the main study (nine documents, six per arm, 1,670 questions), the planned and verified pipeline on a 4-bit 7B model reached a quality index of 0.96 (95 % CI 0.93–0.98), against 0.64 for naive RAG, 0.83 for the whole document in one prompt to the same model and 0.95 for the whole document to a large server model, which reached 0.99 with our pipeline. Planning added 0.24 and verification 0.08; another-family verifier and a concept graph did not help; the verifier’s flag agrees only weakly with the judge ($\kappa \leq 0.22$); and with six documents no difference survives Holm correction.

Index Terms—automatic question generation, retrieval-augmented generation, verification, small language models, privacy, LLM-as-a-judge, educational technology, Hungarian

I. INTRODUCTION

Writing good multiple-choice questions is slow: the stem must be unambiguous, exactly one option must be correct according to the course material, and the distractors must be plausible but provably wrong. Large language models (LLMs) can draft such questions [1], [2], and the most common way teachers use them is direct: upload the lecture notes to a chat assistant and ask for “ten multiple-choice questions”. This is convenient, but it has four weaknesses that matter in a school.

(P1) Correctness is not checked. Nothing verifies that the keyed option is supported by the document or that the distractors are actually false; LLMs produce plausible but unsupported statements, which self-verification can reduce but not remove [3], [4]. **(P2) Structure and coverage are not controlled.** The requested number, types and difficulty of questions are only suggested in the prompt, and long inputs

are used unevenly—models attend to the beginning and end of long contexts more than to the middle [5]—so questions cluster on a few sections. **(P3) Privacy.** Teaching material may contain personal data, and a hosted assistant makes the provider a recipient of it. **(P4) Dependence on large hosted models.** Whole documents need long contexts and large models; the small models a school can run itself have limited context and memory.

MIMIR addresses these with a method rather than a prompt, and runs it on a single consumer GPU. We ask three research questions:

- RQ1** Does structured generation—planning, per-question grounding and verification—produce more valid and better-covering exams than prompting a model with the document?
- RQ2** What does each component contribute, and what does it cost in time and model calls?
- RQ3** How close does a 7B model on an 8 GB consumer GPU come to a large server-hosted model?

Contributions. (1) The MIMIR generation method: semantic chunking, a coverage-driven exam blueprint, cited per-question generation and a verifier chain (Section IV). (2) A privacy-preserving local deployment with zero retention of uploaded content and a measured GPU footprint that fits an 8 GB card (Section V). (3) An evaluation methodology and open harness: fourteen single-factor arms, a bilingual dataset with reference questions, a cross-family judge and document-level statistics (Section VI). (4) Results of a pilot, which exposed measurement defects, a malformed fallback and an answer-leakage failure mode, and of a 14-arm main study on nine documents (Section VII).

II. RELATED WORK

Question and distractor generation. Surveys of educational question generation [1] and distractor generation [2] note that evaluation often relies on reference-overlap metrics or small human studies that rarely verify answer correctness. Small models can produce comparable educational questions after further pre-training and fine-tuning [6], [7], and small private models have been compared with large ones for Bloom-aligned assessment design [8]. We use off-the-shelf models without fine-tuning and add explicit grounding and verification.

TABLE I
DESIGN GOALS: SENDING THE DOCUMENT TO A CHAT LLM VS. MIMIR.

Goal	Document → chat LLM	MIMIR
G1 Verifiable	Key not checked; no source reference	Every question cites chunks; grounding and blind-answer checks; unverified questions flagged
G2 Controlled	Count, types and difficulty only requested	Blueprint fixes count, types and Bloom levels; coverage-driven concept choice
G3 Private	Material sent to the provider	Local model; content deleted after each job; optional local-only mode
G4 Feasible	Long context, large model	Short calls within an 8,192–16,384-token context; 7B model on 8 GB
G5 Overseen	Output used as is	Mandatory review in an editor with the cited source; AI disclosure in the export

Retrieval and long contexts. Retrieval-augmented generation (RAG) grounds output in retrieved passages [9]; hybrid retrieval fuses BM25 [10] with dense embeddings [11] using reciprocal rank fusion [12], and graph-based RAG builds a concept graph for questions that need global context [13]. Long-context prompting, the alternative, uses the middle of the input poorly [5].

Verification and small models. Iterative self-feedback [3] and verification questions [4] reduce factual errors, and spending inference compute can substitute for model size [14]. Quantised 4-bit models lose little accuracy on general benchmarks [15], [16], which makes local deployment plausible.

LLM judges and languages. LLM judges correlate with humans [17] but favour their own family [18]. For Hungarian, benchmarks [19] and QA data [20] exist, but we know of no evaluation of exam generation.

III. DESIGN GOALS

Table I contrasts the direct approach with MIMIR on five goals derived from P1–P4 and from the human-oversight requirement of the EU AI Act [21].

IV. THE MIMIR METHOD

MIMIR is a set of services behind a gateway with a web frontend. Fig. 1 contrasts its generation pipeline with the naive RAG baseline, which retrieves the three chunks most similar to the request and asks for the whole exam in one call.

Document processing. Text is extracted from PDF, DOCX or plain text and split into semantic chunks: sentences are embedded with multilingual E5 [11], and a cut is placed where the cosine distance between neighbouring sentences exceeds a percentile (or $\text{mean} + k\sigma$) threshold, within a target size. The encoder works in windows of at most 512 tokens (Section VI-A). Chunks are indexed in a vector store that exists only for the current job.

Blueprint. One call proposes $n + 3$ concepts that cover the document, each with the chunk identifiers that support it.

TABLE II
VERIFIER CHECKS, IN EXECUTION ORDER.

Check	LLM	Feedback on failure
Shape	–	MCQ: 4 options, 1 key
Meta-reference	–	Rephrase without “according to the text”
Citations	–	Cite a chunk from the given context
Grounding and distractors	1	Key not supported by the cited chunks, or a distractor may be true
Blind answer	1	A solver without the key chose another option
Leakage (new)	–	Key contained in the stem, or stem not a question
Duplicate	–	Too similar to an accepted question

Slots are filled greedily, preferring concepts that touch chunks no slot covers yet, so the exam spreads over the document; the three spare concepts serve as replacements. Each slot receives a question type and a Bloom level [22] by largest-remainder allocation from a difficulty-dependent mix (easy: remember/understand; medium: understand/apply/analyse; hard: apply/analyse/evaluate). Count and types are therefore guaranteed by construction (G2).

Grounded generation. For each slot, the concept’s chunks are combined with the top- k retrieved passages (dense, or BM25 and dense fused with RRF), and the model writes *one* question that must cite the chunk identifiers supporting its key (G1). A concept-graph variant (E3) extracts concepts and typed relations per 3,000-character batch, replaces the planner call with graph concepts, offers sibling concepts as distractor ideas, and adds 1-hop neighbour chunks to the context.

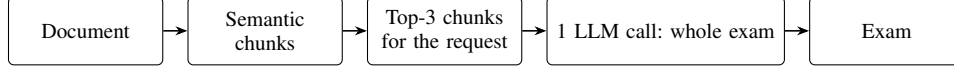
Verification. Checks run cheapest first (Table II). A failed check returns its reason to the generator for one retry; after that the slot moves to a spare concept, and as a last resort the best attempt is kept and flagged as unverified for the teacher. Near-duplicates (embedding similarity > 0.9) are regenerated. After the pilot we added leakage checks, a repair call for malformed last-resort questions, and an optional verifier from another model family.

Human review. The draft opens in an editor where the teacher can edit, reorder, change the key, regenerate a single question and view each question’s cited source passage. Nothing is exported without approval, and the exported PDF states that the content was drafted with an AI system and by which model (G5).

V. PRIVACY AND LOCAL DEPLOYMENT

Zero retention (G3). Uploaded bytes, extracted text, chunks, embeddings and prompts exist only in service memory for the duration of a job; the vector store is cleared when the job ends or fails, and job results expire after 60 minutes. The audit log keeps the job identifier, model, prompt version, quality score and SHA-256 fingerprints and lengths of the request and context—never content. A `LOCAL_ONLY` switch disables every external model call; otherwise the upload screen names where the document will be processed and lets the teacher choose the local model.

E0 – naive RAG (baseline)



E1–E3 – MIMIR blueprint

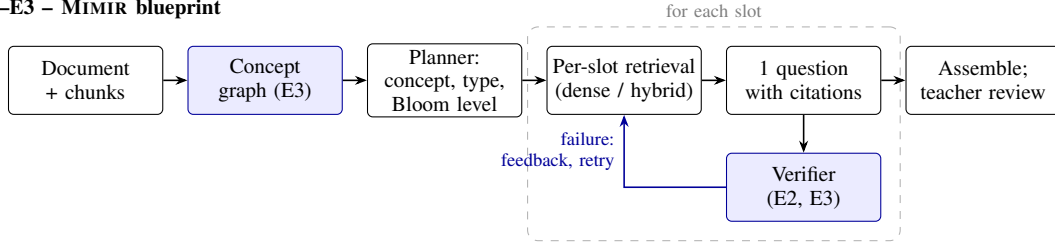


Fig. 1. Naive RAG versus the MIMIR blueprint pipeline. Blue: components switched on and off by the experiment arms. The blueprint fixes the exam structure before any question is written and verifies each question against its own cited context.

TABLE III

ESTIMATED GPU MEMORY AT AN 8,192-TOKEN CONTEXT (FP16 KEY-VALUE CACHE, ≈ 0.6 GB RUNTIME). WEIGHTS: GGUF FILE SIZE; CACHE: FROM THE PUBLISHED ARCHITECTURES.

Model	Quant.	Weights	Cache/token	Total	Card
Qwen2.5-3B	Q4_K_M	1.9 GB	36 KiB	2.8 GB	8 GB
Qwen2.5-7B	Q4_K_M	4.7 GB	56 KiB	5.8 GB	8 GB
Llama-3.1-8B [24]	Q4_K_M	4.9 GB	128 KiB	6.6 GB	8 GB
Qwen2.5-7B	Q8_0	8.1 GB	56 KiB	9.2 GB	12 GB
Qwen2.5-14B	Q4_K_M	9.0 GB	192 KiB	11.2 GB	12 GB

Consumer GPU (G4). The generator is Qwen2.5-7B [23] with 4-bit (Q4_K_M) weights in the Ollama runtime, the only thing on the GPU; embeddings and all services run on the CPU. Because every call in the blueprint pipeline is short, the context stays at 8,192–16,384 tokens; the 7B model’s grouped-query attention needs 56 KiB of key–value cache per token, about 0.9 GB at 16,384 tokens. In the pilot, GPU memory rose from 1.6–1.9 GB (desktop, model not yet loaded) to a peak of 7.0–7.8 GB during generation, so the model and runtime occupied 5.4–5.9 GB—within the budget of an 8 GB card, including the desktop. Table III estimates the budget for other local models: 3–8B models at 4 bits fit 8 GB; a 4-bit 14B model or an 8-bit 7B model needs 12 GB.

VI. EVALUATION METHODOLOGY

The harness (`tests/eval`) runs in three steps: *run* generates exams for every document $\times$ seed of an arm, *score* computes metrics and calls the judge, and *report* produces tables, tests and figures. Each run stores its configuration, git commit and dataset version.

Dataset. Table IV summarises the 39 documents, frozen as `eval-v1`. EduQG [25] and MILQA [20] importers select short, medium and long documents and keep reference questions whose evidence sentence occurs in the document; references are used for retrieval recall and similarity, not as targets.

Arms. Each arm changes one factor (Table V). All local arms use Qwen2.5-7B (Q4_K_M, temperature 0.2); the server

TABLE IV

EVALUATION DATASET `eval-v1`.

Source	Lang	Docs	S/M/L	Median chars	Ref. Q
Own teaching texts	HU	3	3/0/0	1,923	12
OpenStax via EduQG	EN	12	4/4/4	16,752	74
Wikipedia via MILQA	HU	24	8/8/8	8,072	240
Total		39	15/12/12	7,919	326

TABLE V
EXPERIMENT ARMS.

Arm	Configuration
B-doc-L / S	Whole document in one prompt, local 7B / server model (status quo)
E0	Naive RAG: top-3 chunks, whole exam in one call (baseline)
E1	Blueprint: planner, per-slot retrieval, one question per call
E2	E1 + verifier chain with retries
E2x	E2 with a verifier from another family (Llama-3.1-8B)
E2f / E2h	E2 without blind answer test / with hybrid retrieval
E3	E2 + concept graph
E4 / E4b	E0 / E2 with the large server model
E5a / b / c	E0 with the legacy 512-token encoder / mean+1.5 σ cuts / fixed 800-char chunks

arms use a large Qwen-family model on the university GPU server.

Measures. *Format compliance* is 1 if an exam has exactly the requested number and shape of questions. Per question, the judge decides *grounding* (the source supports the key, with a quoted sentence), *distractor validity* (each distractor false and on topic) and *blind answerability* (answering from the source without the key picks the keyed option), and gives 1–5 rubric scores. The *quality index* is the mean of these four rates. *Coverage* is the share of the document’s chunks that are the closest chunk (by embedding) of at least one question; we also record duplicates, meta-references, answer leakage, unverified questions, calls, tokens, time, peak GPU memory, the GPU model and the share of the model held in GPU memory.

Judge and teachers. The judge is gpt-oss-120b [26], from a different model family than the generators, to avoid self-preference [18]. It sees the chunks each question was written from or, for whole-document arms, the first 24,000 characters of the document. Blind teacher rating sheets use the same rubric; Krippendorff’s α [27] and Spearman’s ρ validate the judge.

Statistics. Documents are the unit of analysis: measures are averaged over questions and seeds per document, with 10,000-resample bootstrap CIs [28], paired Wilcoxon tests [29] with Holm correction [30], and rank-biserial effect sizes [31]. Fewer than six documents cannot reach $p < 0.05$, so single-document results are reported descriptively.

A. Measurement validity

Building the harness exposed two defects that had made earlier end-to-end measurements invalid. *Fallback exams counted as results:* when every model call failed, the backend returned a hard-coded one-question exam, which a test run on 3 May 2026 reported as three documents generated in 4–9 s; it is now counted as a failure. *Truncated chunking:* the chunker embedded each document in one 512-token pass, so every sentence after roughly the first page received a zero vector—an estimated 68–71 % of the average eval-v1 document. The encoder now works in windows; arm E5a keeps the old one. A third issue, answer leakage, was found in the pilot.

VII. RESULTS

A. Pilot

The pilot ran ten arms on one short Hungarian text (1,702 characters, 4 chunks) with three seeds, on a consumer GPU (NVIDIA GeForce RTX 3070 Ti, 8 GB). Four findings changed the main study. *Noise:* E0 and E5a received identical chunks, context and seeds yet scored 0.83 and 0.90, so on one document differences below about 0.07 are noise. *Malformed fallback:* in one exam of every verifier arm a kept last-resort question had only two or three options, which zeroed that exam’s format compliance; verification then lowered the index (E1 0.88, E2 0.76). *Self-verification at chance level:* questions the 7B verifier passed were judged grounded less often (66 %) than those it flagged (75 %; κ from -0.14 to 0.07). *Answer leakage:* the verifier’s feedback loop pushed the small model towards copying a source sentence into the stem and repeating it as the key, the cheapest way to pass a grounding check. In 20 % of the concept-graph arm’s questions the key’s words appeared in the stem (≥ 80 % token overlap)—e.g. a stem quoting that coffee reached Europe in the 17th century, keyed with the same sentence and distractors naming other centuries—and the judge rated such items 5 of 5. The main study therefore keeps only well-formed last-resort questions (with one repair call), rejects stems that contain the key or are not questions (Table II), and reports leakage.

B. Main study

Setup. The main study ran on 27–28 September 2026 with the corrected pipeline and one retry per slot. It covers nine

TABLE VI
MAIN STUDY. INDEX: QUALITY INDEX (MEAN OF DOCUMENT MEANS) WITH 95 % BOOTSTRAP CI; RATES IN %. ALL ARMS BUT E4 AND E4B RAN ON THE SAME SIX DOCUMENTS (3 HU, 3 EN); E4 AND E4B ON SIX EN DOCUMENTS, THREE OF THEM SHARED WITH THE OTHERS.

Arm	Index	95 % CI	Form.	Grnd.	Blind	Dist.	Cov.	Leak.	Unv.	Calls	Min
B-doc-L	0.83	0.76–0.90	83	79	86	85	45	1	–	1	0.4
B-doc-S	0.95	0.87–0.99	100	90	100	89	47	0	–	1	4.0
E0	0.64	0.46–0.81	100	43	62	52	25	0	–	1	0.3
E1	0.88	0.82–0.94	100	80	80	91	45	8	1.2	17	1.9
E2	0.96	0.93–0.98	100	90	97	96	44	3	4.8	51	3.5
E2x	0.89	0.83–0.95	100	78	83	95	42	2	2.5	19	7.6
E2f	0.90	0.84–0.95	100	78	88	92	44	3	3.7	38	2.8
E2h	0.92	0.90–0.94	100	82	92	94	42	2	5.2	53	3.7
E3	0.88	0.79–0.95	100	83	87	83	40	0	4.8	56	5.5
E4	0.91	0.80–0.98	100	77	98	87	9	0	–	1	5.3
E4b	0.99	0.97–1.00	100	98	100	98	28	0	0.0	37	47.7
E5a	0.87	0.81–0.92	100	74	84	90	36	2	–	1	0.3
E5b	0.70	0.54–0.85	100	39	78	63	35	1	–	1	0.3
E5c	0.86	0.70–0.97	100	82	82	81	51	2	–	1	0.3

Form. format compliance; Grnd. grounding; Dist. distractor validity; Cov. coverage; Leak. key in stem; Unv. unverified questions per exam; Calls: mean model calls per exam; Min: median minutes per exam.

TABLE VII
MAIN STUDY: PAIRED WILCOXON TESTS ON THE PER-DOCUMENT QUALITY INDEX. n : SHARED DOCUMENTS; r : RANK-BISERIAL EFFECT SIZE ($r > 0$: NEW ARM BETTER); p HOLM-ADJUSTED OVER THE SIX COMPARISONS.

Comparison	n	Ref.	New	Δ	r	p_{Holm}
E1 vs B-doc-L	6	0.83	0.88	+0.04	0.24	0.875
E1 vs E0	6	0.64	0.88	+0.24	0.62	0.875
E2 vs E1	6	0.88	0.96	+0.08	1.00	0.188
E2x vs E2	6	0.96	0.89	−0.06	−0.90	0.312
E3 vs E2	6	0.96	0.88	−0.07	−0.62	0.875
E4 vs E0	3	0.40	0.93	+0.54	1.00	0.875

of the 39 documents, six per arm. Twelve arms—all local arms and B-doc-S—ran on the three short Hungarian teaching texts (1,696–2,295 characters) and a short, a medium and a long OpenStax text (6,000, 24,937 and 68,063 characters); E4 and E4b ran on six English OpenStax documents, three of them shared with the others. Single-call arms used three seeds, planned arms one: 1,670 questions, all judged; teacher ratings are not yet in. With six documents, $p < 0.05$ requires all six differences to share a sign.

Method versus prompting (RQ1). On the shared six documents, the planned and verified pipeline on the local 7B model (E2) scored 0.96 (CI 0.93–0.98), against 0.64 for naive RAG (E0) and 0.83 for the whole document in one prompt to the same model (B-doc-L). B-doc-L failed the format in 17 % of its exams and saw only the first 30,000 characters of the longest document. Given the whole document, the server model (B-doc-S) reached 0.95, as high as the local E2. Coverage did not separate the method from whole-document prompting (B-doc-L 45 %, B-doc-S 47 %, planned arms 40–45 %); the clustering seen in the pilot reappeared only with naive RAG (E0 25 %, E4 9 %).

Components (RQ2). Planning (E1) raised the index of naive RAG by 0.24 ($r = 0.62$), grounding from 43 to 80 %, distractor validity from 52 to 91 % and coverage from 25 to

45 %, at 17 calls and 1.9 minutes per exam—but only on the English documents (0.40→0.91; Hungarian 0.88→0.85). Unlike in the pilot, verification (E2) improved on planning on every one of the six documents (+0.08, $r = 1.00$, unadjusted $p = 0.031$, Holm 0.19): grounding rose from 80 to 90 % and blind answerability from 80 to 97 %, at 51 calls and 3.5 minutes per exam, and every verifier arm now had 100 % format compliance. Dropping the blind-answer test (E2f, 0.90) or adding hybrid retrieval (E2h, 0.92) did not help. The Llama-3.1-8B verifier (E2x) passed 75 % of the questions instead of 52 % and lowered the index to 0.89 ($r = -0.90$); the concept graph (E3) lowered it to 0.88, mainly through weaker distractors (83 vs. 96 %).

The verifier as a classifier. The pilot’s direction reversed: of 360 questions in the six verifier arms, those the verifier passed ($n = 234$) were judged grounded more often (89 %) than those it flagged ($n = 126$, 77 %), and the judge would use 85 vs. 68 % of them; without E4b, whose server verifier passed all 60 questions, 150 of the 174 passed local questions (86 %) were grounded. Agreement stayed weak, with κ from 0.02 to 0.22 for the local verifiers: precision 0.79–0.94 but recall 0.54–0.79, so about three quarters of the flagged questions are in fact sound. The flag is a useful warning, not a verdict.

Leakage and degenerate questions. The rule checks removed most leakage from the verifier arms (key in stem 0–3 %; E3 0 % against 20 % in the pilot); statement stems are now most frequent with fixed-size chunks (E5c, 41 %) and in B-doc-L (17 %). A new failure appeared with naive RAG on the server model: E4 asked about the retrieved text itself (“How many words are in Link to Learning?”), and the judge rated such items 5 of 5. They inflate E4’s index (0.91) and fit its 9 % coverage.

Naive RAG is fragile. The chunking arms show why E0 was weakest. On the Hungarian texts the semantic variants scored 0.87–0.89, but on the English documents E0 and E5b collapsed (0.40 and 0.51, grounding 0 %), while the legacy encoder (E5a, 0.87) and fixed 800-character chunks (E5c, 0.96) did not. OpenStax texts contain short structural lines (“Link to Learning”) that semantic chunking, having no minimum chunk size, turns into chunks of their own, and the naive query—a task description such as “10 multiple-choice questions”—lands on them: the judge saw contexts that were a heading or no words. Retrieval itself was sound (recall@3 of the reference evidence 89–95 % in every arm), and the planned pipeline, which searches per concept, was unaffected (E1 0.91 on the English documents).

Server versus local (RQ3). On the three documents shared by all arms, the local E2 scored 0.95 and the server model with naive RAG (E4) 0.93; naive RAG scored 0.40 locally on them. The planned and verified pipeline on the server model (E4b) reached the study’s highest index, 0.99, on its six English documents—the method carries over to a larger model—but needed a median of 47.7 minutes per exam on the shared server, against 3.5 for E2 on the local GPU.

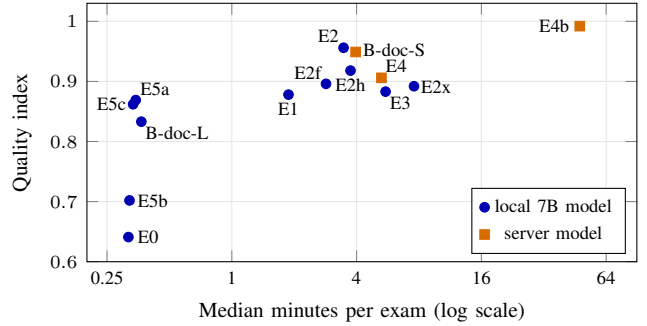


Fig. 2. Main study: quality versus time per exam. Blue: local 7B model; orange: large server model on the shared university server. E4 and E4b ran on partly different documents.

VIII. DISCUSSION

Why a method rather than a prompt. The same 7B model went from 0.64 (naive RAG) and 0.83 (whole document) to 0.96 once the exam was planned and verified; naive RAG depended on chunking details that a per-concept search avoids; and per-question citations let the judge—the teacher, in practice—check each key, which exposed wrong keys, leakage and degenerate questions. On these mostly short documents, however, the whole-document server baseline was as good and covered as much, so the case against hosted models rests on verifiability, privacy and local operation (P1, P3, P4), and the coverage argument (P2) holds against naive RAG only. We recommend E2 as the local configuration, with the verifier’s flag shown as a warning.

What the main study does not show yet. It covers nine of the 39 documents, six per arm; its Hungarian part is three short texts and no Wikipedia article, so language is confounded with source and length; E4 and E4b ran on partly different documents; planned arms used one seed; and no comparison survives Holm correction. The judge is not yet validated by the teachers (*teacher ratings: Krippendorff’s α and judge–teacher agreement*). The next run puts the core arms on all 39 documents with identical document sets, gives the judge the full document for whole-document arms, and measures a minimum chunk size for naive RAG.

Reproducibility. Every run stores its configuration, git commit, dataset version and generator settings; judge answers are cached, so re-scoring is deterministic; the run→score→report flow is covered by end-to-end tests with a mock LLM; and the report writes Tables VI and VII. The pilot’s raw outputs and both reports are released with the code.

Limitations. The judge sees each question’s cited context and may miss contradictions elsewhere; it rated leaking and degenerate questions highly. For whole-document arms it sees at most the first 24,000 characters: on the 68,063-character law document it judged B-doc-S’s question on the “access economy” unsupported because the term first appears at character 38,603, so whole-document grounding on long documents is a lower bound. Only multiple-choice questions are evaluated. Run times (local GPU of the main run: *model*; a shared server)

are indicative only.

IX. CONCLUSION

MIMIR turns exam generation from a single prompt into a verifiable method that runs on a consumer GPU without retaining teaching material. In a main study on nine documents, the planned and verified pipeline on a local 7B model scored 0.96, against 0.64 for naive RAG and 0.83 for the whole document in one prompt to the same model, and matched a large server model given the whole document (0.95); on the server model the same pipeline reached 0.99. Planning and verification helped, a verifier from another family and a concept graph did not, and the verifier’s per-question flag agrees only weakly with the judge. Building the evaluation exposed failure modes that inflate automatic and LLM-judge scores. The full 39-document run with teacher ratings will test these findings; code, configurations, dataset manifests and reports are released.

ETHICAL CONSIDERATIONS

Uploaded documents are deleted after each job and no student data is processed. Every generated question requires teacher review, and exports disclose AI assistance and the model used, in line with the EU AI Act [21]. Evaluation documents are the authors’ own or openly licensed (CC BY 4.0, CC BY-SA 4.0).

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[To be completed: funding, access to the university GPU server, teacher raters.]

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